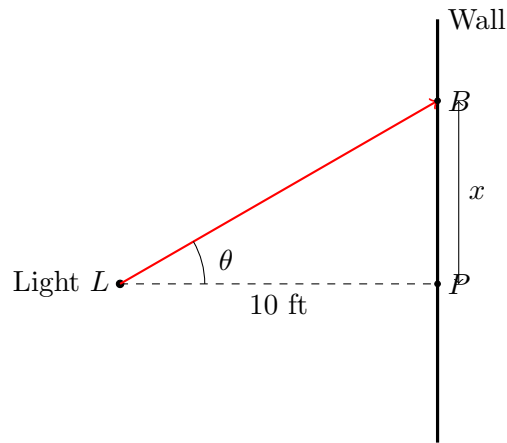


extra credit C

my uni: sn3136

Exercise 1: Rotating Searchlight

(i) Diagram



so we have θ which is the angle between the beam and the perpendicular from the light to the wall, and we got x which in here is the distance along the wall from P (the foot of the perpendicular) to the spot of light B

(ii)

the searchlight rotates at 2 revolutions per minute so

$$\frac{d\theta}{dt} = 2 \cdot 2\pi = 4\pi \text{ rad/min}$$

from the right triangle LPB

$$\tan \theta = \frac{x}{10} \implies x = 10 \tan \theta$$

after differentiating we get:

$$\boxed{\frac{dx}{dt} = 10 \sec^2 \theta \frac{d\theta}{dt} = 40\pi \sec^2 \theta \text{ ft/min}}$$

(iii)

(a) when the beam is perpendicular to the wall it points straight at P , so $\theta = 0$ and $\sec^2 \theta = 1$:

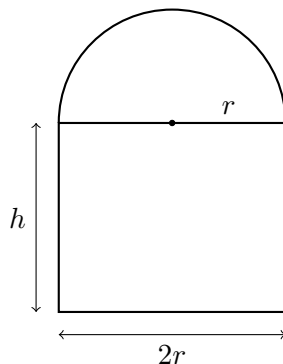
$$\left. \frac{dx}{dt} \right|_{\theta=0} = 40\pi(1) = 40\pi \text{ ft/min} \approx 125.66 \text{ ft/min}$$

(b) when the beam makes a 60° angle with the wall then the angle it makes with the perpendicular LP is $\theta = 90^\circ - 60^\circ = 30^\circ$, so $\sec^2 \theta = \left(\frac{2}{\sqrt{3}}\right)^2 = \frac{4}{3}$:

$$\left. \frac{dx}{dt} \right|_{\theta=30^\circ} = 40\pi \cdot \frac{4}{3} = \frac{160\pi}{3} \text{ ft/min} \approx 167.55 \text{ ft/min}$$

ex 2

(i)



so the rectangle has height h and width $2r$ where r is the radius of the semicircle

(ii)

there's a perimeter constraint: the window's perimeter is the bottom, the two vertical sides, and the semicircular arc, so:

$$2r + 2h + \pi r = 25 \implies h = \frac{25 - 2r - \pi r}{2}$$

we got the area to maximize:

$$A(r) = 2rh + \frac{1}{2}\pi r^2$$

substituting the constraint

$$A(r) = 2r \cdot \frac{25 - 2r - \pi r}{2} + \frac{1}{2}\pi r^2 = 25r - 2r^2 - \pi r^2 + \frac{\pi}{2}r^2 = 25r - 2r^2 - \frac{\pi}{2}r^2$$

and we need $r > 0$ and $h \geq 0$, so basically, $0 < r \leq \frac{25}{2 + \pi}$

(iii)

so then I differentiate and set equal to zero:

$$A'(r) = 25 - 4r - \pi r = 0 \implies r = \frac{25}{4 + \pi}$$

since $A''(r) = -(4 + \pi) < 0$, this is a maximum so then

$$h = \frac{25 - (2 + \pi) \cdot \frac{25}{4 + \pi}}{2} = \frac{25(1 - \frac{2 + \pi}{4 + \pi})}{2} = \frac{25 \cdot \frac{2}{4 + \pi}}{2} = \frac{25}{4 + \pi}$$

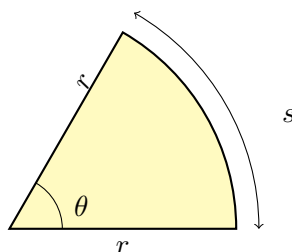
so $h = r = \frac{25}{4 + \pi}$ and the rectangle's width is $2r = \frac{50}{4 + \pi}$

$\text{Width} = \frac{50}{4 + \pi} \text{ ft} \approx 7.00 \text{ ft}, \quad \text{Height} = \frac{25}{4 + \pi} \text{ ft} \approx 3.50 \text{ ft}$

so the maximum area is $A = \frac{625}{2(4 + \pi)} \approx 43.75 \text{ ft}^2$

ex 3

(i)



so we have r which is the radius of the pizza and s the arc length of the slice (with central angle θ , where $s = r\theta$)

(ii)

perimetre constraint: the slice's perimeter has two radii plus the arc:

$$2r + s = 20 \implies s = 20 - 2r$$

we need $r > 0$ and $s \geq 0$ so $0 < r \leq 10$

area to maximize is:

$$A = \frac{1}{2} r s = \frac{1}{2} r (20 - 2r) = 10r - r^2.$$

(iii)

$$A'(r) = 10 - 2r = 0 \implies r = 5.$$

so since $A''(r) = -2 < 0$, this is a maximum and then $s = 20 - 2(5) = 10$, and the central angle is

$$\theta = \frac{s}{r} = \frac{10}{5} = 2 \text{ rad } (\approx 114.6^\circ)$$

Diameter of pizza = $2r = 10 \text{ in}$
--

the fraction of the whole pizza eaten:

$$\frac{A_{\text{slice}}}{A_{\text{pizza}}} = \frac{10(5) - 5^2}{\pi(5)^2} = \frac{25}{25\pi} = \frac{1}{\pi} \approx 0.318$$

so the slice is $\frac{1}{\pi}$ ($\approx 31.8\%$) of the whole pizza

ex4, sketch of $f(x) = \frac{2-x^2}{3-2x}$

so we got $f'(x) = \frac{2(x-1)(x-2)}{(3-2x)^2}$

(i)

domain: $3-2x \neq 0 \iff x \neq \frac{3}{2}$. so

$$\text{Dom}(f) = (-\infty, \frac{3}{2}) \cup (\frac{3}{2}, \infty)$$

x-intercepts: $2-x^2 = 0 \implies x = \pm\sqrt{2}$, so we got the points $(\sqrt{2}, 0)$ and $(-\sqrt{2}, 0)$

y-intercept: $f(0) = \frac{2}{3}$, so we got the point $(0, \frac{2}{3})$.

(ii)

vertical asymptote: at $x = \frac{3}{2}$ so at $x = \frac{3}{2}$ the numerator is $2 - \frac{9}{4} = -\frac{1}{4} \neq 0$ so:

$$\lim_{x \rightarrow (3/2)^-} f(x) = \frac{-1/4}{0^+} = -\infty, \quad \lim_{x \rightarrow (3/2)^+} f(x) = \frac{-1/4}{0^-} = +\infty$$

slant asymptote is so the polynomial long division of $-x^2 + 2$ by $-2x + 3$ gives

$$\frac{2-x^2}{3-2x} = \frac{x}{2} + \frac{3}{4} + \frac{1}{4(2x-3)}$$

and since $\frac{1}{4(2x-3)} \rightarrow 0$ as $x \rightarrow \pm\infty$, the slant asymptote is

$$\boxed{y = \frac{x}{2} + \frac{3}{4}}$$

so

$$\lim_{x \rightarrow \infty} f(x) = +\infty, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

there's no horizontal asymptotes

(iii)

$f'(x) = 0$ when the numerator $2(x-1)(x-2) = 0$, so $x = 1$ or $x = 2$ so basically f' is never undefined on the domain so then the critical numbers are $x = 1$ and $x = 2$:

$$f(1) = \frac{2-1}{3-2} = 1, \quad f(2) = \frac{2-4}{3-4} = 2$$

so critical points: $(1, 1)$ and $(2, 2)$

(iv)

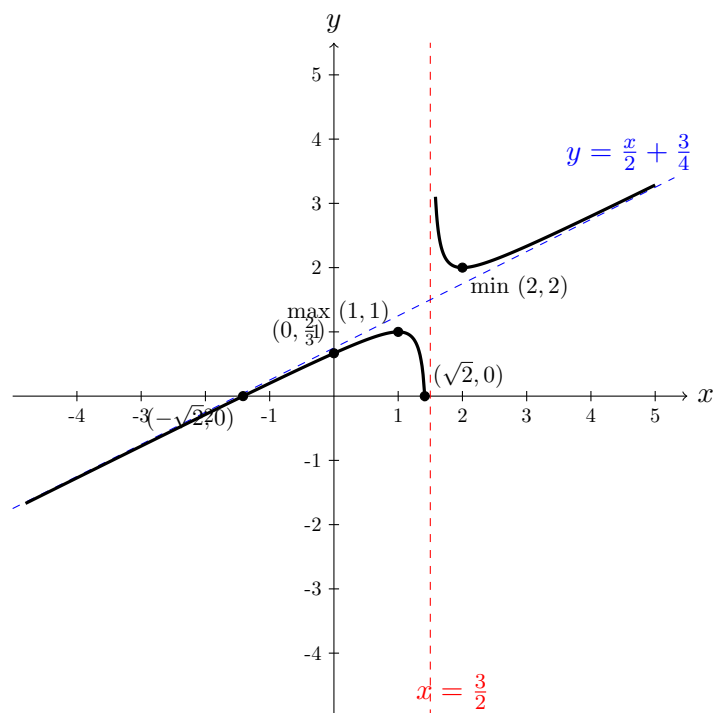
the denominator $(3-2x)^2 > 0$ everywhere on the domain so the sign of f' is the sign of $(x-1)(x-2)$:

Interval	$(-\infty, 1)$	$(1, \frac{3}{2})$	$(\frac{3}{2}, 2)$	$(2, \infty)$
sign of f'	+	-	-	+

at $x = 1$: f' goes from + to -, so $(1, 1)$ is a **local maximum**

At $x = 2$: f' goes from - to +, so $(2, 2)$ is a **local minimum**

(v) my graph



so basically:

- as $x \rightarrow -\infty$, $f \rightarrow -\infty$ along $y = \frac{x}{2} + \frac{3}{4}$ (from below)
- f increases on $(-\infty, 1)$, reaches the local max $(1, 1)$, then decreases on $(1, \frac{3}{2})$ down to $-\infty$
- f comes down from $+\infty$ at $x = \frac{3}{2}^+$, decreases on $(\frac{3}{2}, 2)$ to the local min $(2, 2)$, then increases on $(2, \infty)$ approaching $y = \frac{x}{2} + \frac{3}{4}$ from above